

ABSTRACT

The diagnosis of breast cancer is a critical process that determines the course of treatment and ultimately, patient outcomes. However, traditional methods employed in this process have been plagued by inconsistencies and variability. This has led to a pressing need for innovative solutions that can provide accurate and reliable diagnosis.

To address this challenge, a novel deep learning model has been developed. This model leverages the prowess of convolutional neural networks (VGG16) and self-attention mechanisms to detect the spread of cancer to lymph nodes with unprecedented accuracy. By incorporating a pre-processing step that refines images to enhance clarity, and a self-attention component that highlights critical areas of the image, the model is able to provide a more accurate diagnosis.

Through a rigorous evaluation process, employing metrics such as accuracy, precision, loss rate, robustness, and F1-score, the efficacy of the model has been demonstrated. The model has the potential to transform the field of breast cancer diagnosis, leading to better patient outcomes and improved quality of life. With its ability to provide accurate and reliable diagnosis, it is poised to make a significant impact in the medical community and beyond.

INTRODUCTION

Breast cancer remains a leading cause of cancer-related deaths among women worldwide, making early detection and accurate staging crucial for improved patient outcomes. A key factor in determining prognosis and treatment strategy is the status of the axillary lymph nodes (ALNs). The presence of metastasis in these nodes strongly indicates disease progression and influences decisions regarding surgery, radiation therapy, and systemic treatment. Traditionally, ALN assessment relies on histopathological examination of biopsy or surgical specimens, a process involving manual review of whole slide images (WSIs) by pathologists. While this remains the gold standard, it suffers from inherent limitations, including subjective interpretation, significant time consumption, and potential inter-observer variability.

These challenges are exacerbated by the increasing volume of histopathology data and the limited availability of expert pathologists, particularly in resource-constrained settings. Consequently, there is a pressing need for automated and reliable solutions for ALN metastasis detection that can assist pathologists, improve diagnostic accuracy, and streamline clinical workflows. The emergence of digital pathology and rapid advancements in deep learning offer a promising avenue for addressing these challenges. Deep learning models, especially convolutional neural networks (CNNs), have demonstrated remarkable success in analyzing complex image data, including histopathological slides.

Furthermore, the incorporation of self-attention mechanisms into these models has shown potential to enhance feature extraction and improve the representation of spatial and contextual information within WSIs. This research proposes DLCNBC-SA, a novel deep learning-based classification model, to predict ALN metastasis status in early-stage breast cancer patients. By leveraging the power of CNNs for feature extraction and incorporating a self-attention mechanism, DLCNBC-SA aims to improve the accuracy and efficiency of ALN metastasis detection, ultimately contributing to better patient care and more informed clinical decision-making.

MOTIVATION

1. **Crucial Role of ALN Status:** Axillary lymph node (ALN) metastasis is a critical prognostic factor and a key determinant of treatment strategies in breast cancer.
2. **Limitations of Manual Assessment:** Traditional histopathological evaluation of WSIs is time-consuming, labor-intensive, and subjective, leading to inter-observer variability and potential diagnostic inconsistencies.
3. **Growing Data Volumes:** The increasing volume of digital pathology data makes manual analysis increasingly challenging and unsustainable.
4. **Need for Automation:** There is a critical need for automated solutions to assist pathologists in efficiently and accurately assessing ALN status.
5. **Potential of Deep Learning:** Deep learning, particularly convolutional neural networks (CNNs), has shown remarkable promise in analyzing complex image data, including histopathological slides.
6. **Enhancing with Self-Attention:** Integrating self-attention mechanisms with CNNs can further improve feature extraction and capture crucial contextual information from WSIs.
7. **Preoperative Prediction Advantage:** A reliable preoperative prediction of ALN metastasis can facilitate more personalized treatment planning and potentially reduce the need for unnecessary surgical procedures.
8. **Improving Patient Outcomes:** Ultimately, this research is driven by the goal of improving patient outcomes through more accurate, timely, and efficient ALN metastasis detection.

PROBLEM STATEMENT

- **Input:**
 - * Whole slide images (WSIs) of breast cancer tissue samples.
 - * Clinical data of early breast cancer (EBC) patients (e.g., age, tumor size, ER/PR/HER2 receptor status).
- **Processing:**
 - * WSI Preprocessing:
 - * Segmentation of WSIs into smaller, manageable patches.

Feature Extraction:

- * Using a pre-trained Convolutional Neural Network (CNN) (e.g., VGG16_BN) to extract deep features from the image patches.
- * Applying a self-attention mechanism to enhance feature maps and capture contextually relevant information.

Clinical Data Processing:

- * Normalization and encoding of clinical parameters.
- * Multi-Modal Feature Fusion:
 - * Concatenating image-derived features with processed clinical data.

Classification:

- * Feeding the fused feature vector into a dense neural network for binary classification (N0 - negative for axillary lymph node metastasis, N(+) - positive for axillary lymph node metastasis).

Segmentation:

- * Segmentation of WSIs into smaller patches (e.g., 256x256 pixels) is performed as a preprocessing step before feature extraction. This is necessary because the high resolution of the WSIs makes it computationally challenging to process the entire image at once.

Output:

- * This output represents the model's prediction of probability of axillary lymph node metastasis based on the provided image and clinical data with regression.

SCOPE OF THE WORK

1. **Data Acquisition and Preprocessing:** Utilizing a large, well-curated dataset from the BCNB comprising WSIs and corresponding clinical data of 1,058 early breast cancer (EBC) patients. This dataset is split into training and independent test cohorts.
2. **Model Development:** Designing a hybrid deep learning model that combines a CNN-based feature extractor with a self-attention mechanism to enhance the capture of both local and global image features.
3. **Integration of Multi-Modal Data:** Combining the features extracted from WSIs with clinical parameters such as age, tumor size, and receptor status to improve diagnostic accuracy.
4. **Experimental Evaluation:** Conducting comprehensive experiments including comparative analysis against state-of-the-art (SOTA) methods and ablation studies to determine the optimal configuration regarding feature extractors, data augmentation techniques, and optimizers.
5. **Performance Metrics:** Focusing on robust evaluation metrics (AUC, accuracy, sensitivity, specificity, etc.) to rigorously validate the model's performance under various experimental conditions.

OBJECTIVES

The primary objectives of the research are to:

1. To Develop an automated diagnostic tool that accurately classifies axillary lymph node (ALN) metastasis status in early breast cancer patients.
2. To Enhance feature extraction by integrating convolutional neural networks with self-attention mechanisms to capture both local details and global contextual information from whole slide images.
3. To Integrate multi-modal data—combining imaging features with clinical parameters such as age, tumor size, and receptor statuses—to improve diagnostic accuracy.
4. To Optimize model performance through comprehensive ablation studies and data augmentation techniques, addressing challenges such as overfitting and class imbalance.

5. To Validate the framework's effectiveness and reliability by benchmarking it against state-of-the-art methods using evaluation metrics (e.g., AUC, accuracy, sensitivity, specificity).
6. To Streamline the diagnostic workflow by reducing interpretation time and minimizing inter-observer variability, ultimately supporting better clinical decision-making.

REVIEW OF LITERATURE

A comprehensive review of the literature reveals several important themes that justify the proposed approach:

“Generalization of Deep Learning in Digital Pathology: Experience in Breast Cancer Metastasis Detection”

This study investigates the challenges of applying AI models in real-world pathology by evaluating the generalization of a deep learning model trained on CAMELYON datasets for detecting breast cancer metastases in lymph nodes. The study tests the model's performance on unseen sentinel lymph node data and axillary lymph node dissections, revealing a significant drop in accuracy, especially for the latter. To address this, the researchers retrain the model using local data from Linköping University Hospital, employing transfer learning, increased sampling rates for local patches, and hard negative mining to reduce false positives. The deep learning model is based on a modified DenseNet architecture, trained and validated using whole-slide images (WSI) with annotated tumor regions. Performance evaluation is conducted using Area Under Curve (AUC) and Free Response Operating Characteristic (FROC) metrics, along with qualitative pathologist reviews. The retrained model shows improved accuracy, particularly for axillary nodes, but still demonstrates the need for further refinements. The study underscores the importance of retraining AI models with diverse datasets to ensure robustness in clinical applications.

“DLCNBC-SA: A Model for Assessing Axillary Lymph Node Metastasis Status in Early Breast Cancer Patients”

This study presents a deep learning-based model (DLCNBC-SA) designed to predict axillary lymph node (ALN) metastasis from whole slide imaging (WSI) and clinical data. The model integrates a convolutional neural network (CNN) with a self-attention mechanism (CNBC-SA) to

enhance feature extraction. Methods include training on core-needle biopsy WSIs from the BCNB dataset (1,058 patients) and using data augmentation (RandomRotation) and the Adadelta optimizer for improved performance. The study evaluates various CNN architectures (e.g., VGG16, ResNet50, EfficientNet) and compares DLCNBC-SA against state-of-the-art models using metrics like area under the curve (AUC), sensitivity, and specificity. Results show that DLCNBC-SA outperforms existing models, achieving an AUC of 0.882. Ablation studies confirm that attention mechanisms and optimal feature extraction significantly improve classification accuracy. The study concludes that the proposed model enhances diagnostic efficiency and could serve as a reliable tool for assisting pathologists in ALN metastasis assessment.

“Predicting Axillary Lymph Node Metastasis in Early Breast Cancer Using Deep Learning on Primary Tumor Biopsy Slides ”

This study presents a deep learning (DL)-based model to predict axillary lymph node (ALN) metastasis in early breast cancer (EBC) using primary tumor biopsy slides. The proposed DL core-needle biopsy (DL-CNB) model utilizes an attention-based multiple-instance learning (AMIL) framework to analyze whole-slide images (WSIs) of biopsy samples. Convolutional neural networks (CNNs) extract deep features, with VGG16_BN proving to be the most effective feature extractor. To enhance prediction accuracy, clinical data is integrated into the model (DL-CNB+C), yielding an AUC of 0.831, with even higher performance in patients under 50 years (AUC: 0.918). Model interpretation highlights nuclear morphometric features such as density, circumference, circularity, and orientation as key indicators of ALN metastasis. This non-invasive, AI-driven method offers a potential alternative to sentinel lymph node biopsy (SLNB), aiding preoperative decision-making and personalized treatment strategies.

“Artificial Intelligence Aided Diagnosis of Breast Cancer Lymph Node Metastasis on Histologic Slides in a Digital Workflow”

This study evaluates an AI-based approach for detecting lymph node (LN) metastasis in invasive breast carcinoma using histologic slides within a digital pathology workflow. The research involved two sentinel LN (SLN) cohorts—a validation cohort (234 SLNs) and a consensus cohort (102 SLNs)—along with a nonsentinel LN (NSLN) cohort (258 LNs). Whole slide images (WSIs) were processed using the Visiopharm Integrator System (VIS) metastasis AI algorithm, which identified metastases with 100% sensitivity and a negative predictive value (NPV) of 100%. The

specificity varied, being 41.5% for the SLN validation cohort and 78.5% for the NSLN cohort. False positives were mainly due to histiocytes, crushed lymphocytes, and germinal centers. Pathologists using VIS AI-annotated slides required significantly less time compared to immunohistochemistry (IHC) slides (0.6 vs. 1.0 minutes, $p = 0.0377$), demonstrating improved efficiency. The study suggests that AI integration into digital pathology workflows can enhance screening accuracy and reduce workload, though manual review remains necessary to address false positives.

SYSTEM ARCHITECTURE

The DLCNBC-SA system architecture is designed as a modular pipeline that processes and integrates multi-modal data effectively. The key components include:

1. Preprocessing Module:

- **WSI Segmentation and Patch Extraction:** Due to the high resolution of WSIs, each image is divided into manageable patches (e.g., 256×256 pixels). This step ensures that the model can process high-resolution data without overwhelming computational resources.

- **Clinical Data Processing:** Clinical information such as age, tumor size, and receptor statuses (ER, PR, HER2) is preprocessed through normalization and encoding, ensuring compatibility with the feature extraction pipeline.

2. Feature Extraction Module:

- **CNN Backbone:** A pre-trained CNN (e.g., VGG16_BN) is employed to extract deep feature maps from image patches. The CNN serves as the primary mechanism for learning hierarchical features from complex histopathological images.

- **Self-Attention Mechanism:** An attention module is integrated to enhance the feature maps by assigning dynamic weights to different regions of the image. This mechanism captures both local details and broader spatial context, allowing the model to focus on diagnostically relevant regions.

3. Feature Fusion and Classification Module:

- **Multi-Modal Fusion:** Features from the CNN are flattened and concatenated with the processed clinical data, forming a comprehensive feature vector that encapsulates both imaging and non-imaging information.

- **Classification Network:** A dense network layer (or series of layers) further processes the fused features, culminating in a binary classification output (N0 vs. N(+)). This classifier is responsible for final decision-making based on the integrated features.

4. Training and Optimization Module:

- **Data Augmentation:** To mitigate overfitting and improve model generalization, various data augmentation techniques—such as random rotation, affine transformations, and color jitter—are applied to the training images.

- **Optimization Strategies:** The model is trained using optimizers like SGD and AdaDelta, with hyperparameters tuned to achieve optimal convergence. The selection of the optimizer is guided by experimental results, ensuring that the model not only converges efficiently but also generalizes well to unseen data.

This architecture is designed to be scalable and adaptable, allowing future integration of additional data modalities or enhancements in feature extraction techniques.

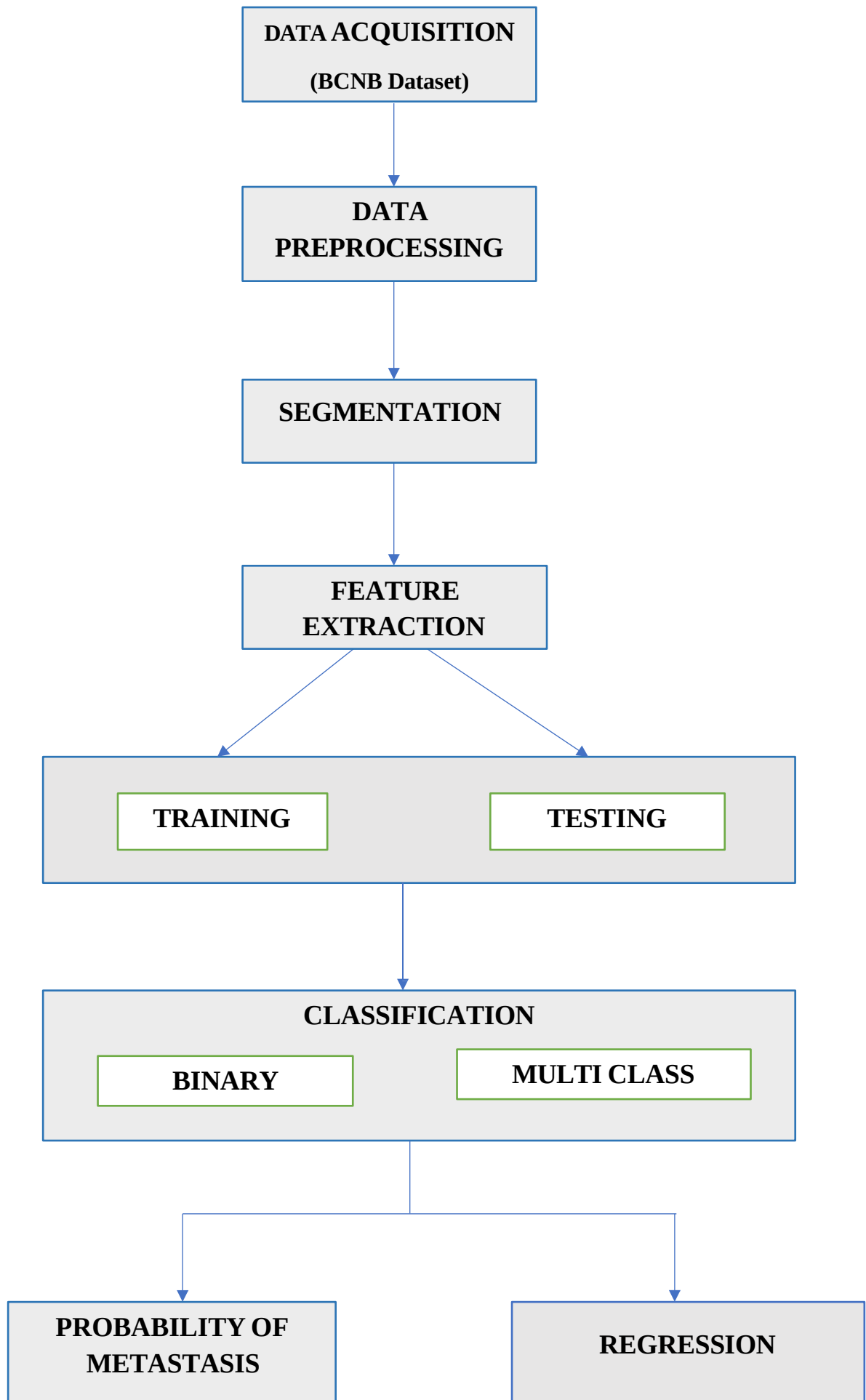


Fig: System Architecture

REFERENCES

1. Ginsburg O, Yip CH, Brooks A, et al. Breast cancer early detection: A phased approach to implementation. *Cancer* 2020;126 Suppl 10:2379–93.
2. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA Cancer J Clin* 2021;71:209–49.
3. Sakai T, Ozkurt E, DeSantis S, et al. National trends of synchronous bilateral breast cancer incidence in the United States. *Breast Cancer Res Treat* 2019;178:161–7.
4. Yan M, Yao J, Zhang X, et al. Machine learning-based model constructed from ultrasound radiomics and clinical features for predicting HER2 status in breast cancer patients with indeterminate (2+) immunohistochemical results. *Cancer Med* 2024;13:e6946.
5. Campanella G, Hanna MG, Geneslaw L, et al. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nat Med* 2019;25:1301–9.
6. Coudray N, Ocampo PS, Sakellaropoulos T, et al. Classification and mutation prediction from non-small cell lung cancer histopathology images using deep learning. *Nat Med* 2018;24:1559–67.
7. Zhang A, Chen Z, Mei S, et al. DLCNBC-SA: a model for assessing axillary lymph node metastasis status in early breast cancer patients. *Quant Imaging Med Surg* 2024;14(8):5831–5844. doi: 10.21037/qims-24-257